EXPERIMENT NUMBER: 2

MATLAB Simulation of Cellular Coverage, Cell Splitting and Handoff in a Multi-Cell Network

Matlab Program

```
% Experiment 2: MATLAB Simulation of Cellular Coverage,
% Cell Splitting and Hand off in a Multi-Cell Network
clear;
clc;
close all;
%% 1. Baseline assumptions
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;

%% 2. Calculate cell area, cell count and capacity
hexArea = @(R) (3*sqrt(3)/2).* R.^2;
cellAreaKm2 = hexArea(radiiKm);
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;

capacityTable = table(radiiKm(:), cellAreaKm2(:), ...
    cellCount(:), repmat(usersPerCell,numel(radiiKm),1), ...
    estimatedUsers(:), ...
    'VariableNames', {'Radius_km','CellArea_km2', ...
    'EstimatedCells','UsersPerCell','IllustrativeUsers'});
disp(capacityTable);

%% 3. Plot a 7-cell hexagonal cluster
R = radiiKm(1);
```

```
centers = [0 0;
    1.5*R sqrt(3)*R/2;
    1.5*R -sqrt(3)*R/2;
    0 sqrt(3)*R;
    0 -sqrt(3)*R;
    -1.5*R sqrt(3)*R/2;
    -1.5*R -sqrt(3)*R/2];
theta = (0:6)*pi/3;
figure;
hold on;
axis equal;
grid on;

for k = 1:size(centers,1)
    plot(centers(k,1) + R*cos(theta), ...
        centers(k,2) + R*sin(theta), 'k-');
text(centers(k,1), centers(k,2), ...
        sprintf('Cell %d',k), ...
        'HorizontalAlignment','center');
end
xlabel('x (km)');
ylabel('y (km)');
title('Idealized Seven-Cell Hexagonal Cluster');
hold off;

%% 4. Simulate mobile movement and handoff
mobileX = linspace(-0.2*R, 1.7*R, 300);
mobileY = zeros(size(mobileX));
d1 = hypot(mobileX, mobileY);
d2 = hypot(mobileX - 1.5*R, mobileY);
```

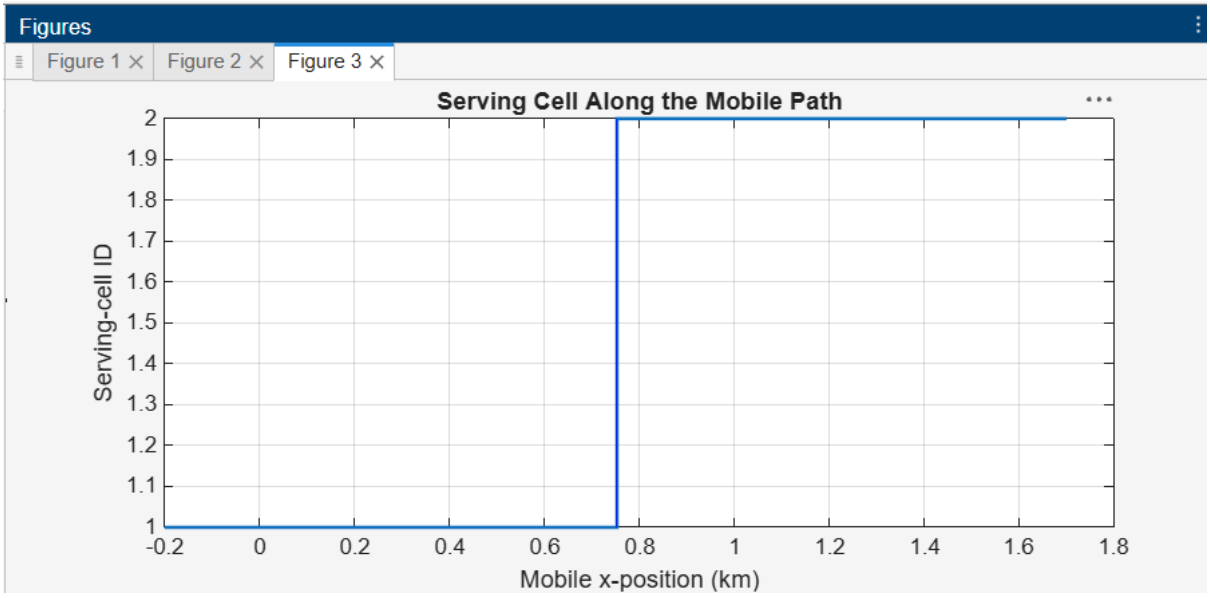
```
servingCell = ones(size(mobileX));
servingCell(d2 < d1) = 2;
handoffIndex = find(diff(servingCell) ~= 0, 1, 'first') + 1;

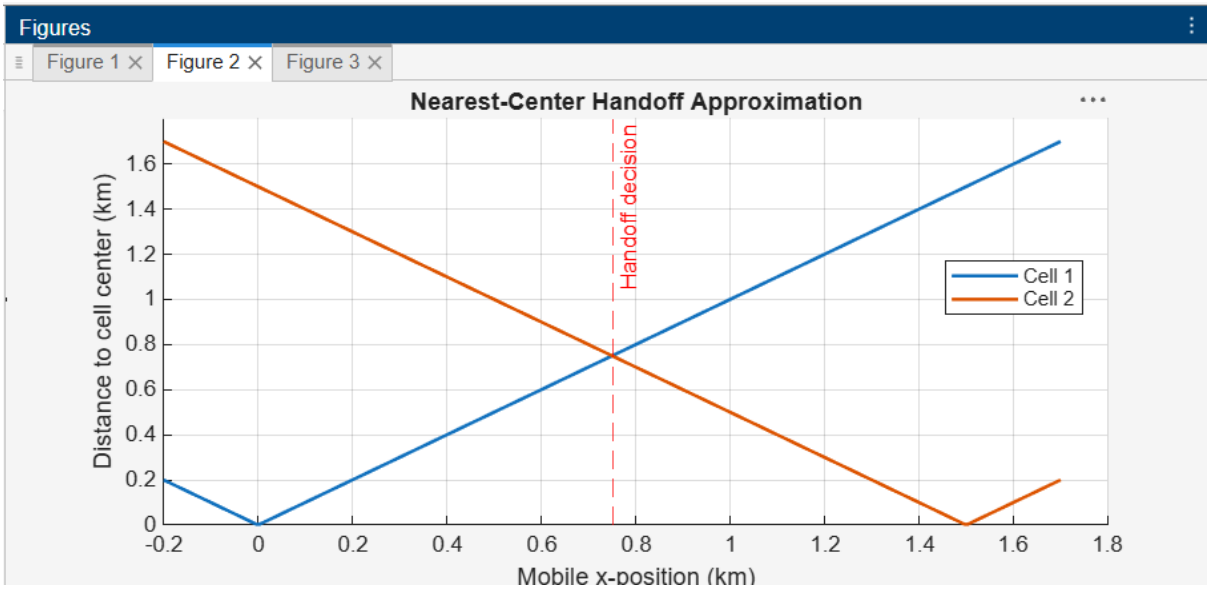
%% 5. Plot nearest-center handoff approximation
figure;
hold on;
grid on;
plot(mobileX, d1, 'LineWidth', 1.5);
plot(mobileX, d2, 'LineWidth', 1.5);
if ~isempty(handoffIndex)
    xline(mobileX(handoffIndex), '--r', 'Handoff decision');
end
xlabel('Mobile x-position (km)');
ylabel('Distance to cell center (km)');
legend('Cell 1','Cell 2','Location','best');
title('Nearest-Center Handoff Approximation');
hold off;

%% 6. Plot serving cell along the mobile path
figure;
stairs(mobileX, servingCell, 'LineWidth', 1.5);
grid on;
xlabel('Mobile x-position (km)');
ylabel('Serving-cell ID');
title('Serving Cell Along the Mobile Path');
xlim([-0.2 1.8]);
ylim([1 2]);
yticks(1:0.1:2);
```

```
if ~isempty(handoffIndex)
    xline(mobileX(handoffIndex), 'b-');
    fprintf('Geometric handoff at x = %.3f km\n', ...
        mobileX(handoffIndex));
end
```

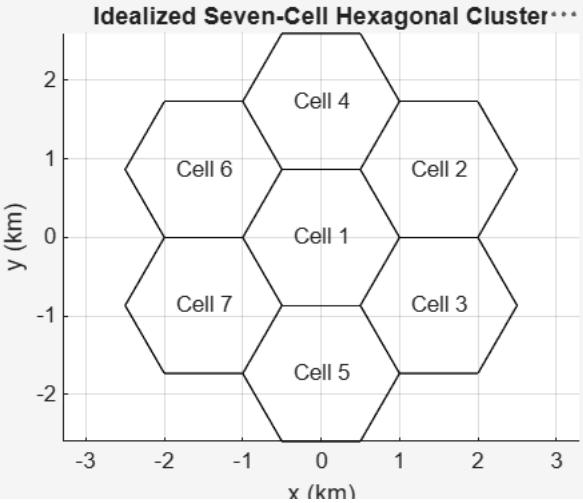
Result





Figures

Figure 1 | Figure 2 | Figure 3



Command Window

Radius_km	CellArea_km2	EstimatedCells	UsersPerCell	IllustrativeUsers
1	2.5981	2	30	60
0.5	0.64952	8	30	240

```
Geometric handoff at x = 0.753 km
>>
>>
```